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EXAMPLE WORKFLOWS AND OPERATING MODES

A. Overview

In various embodiments, the disclosed system supports multiple workflows comprising baseline establishment, longitudinal tracking, geometry acquisition, internal geometry/compliance acquisition, context conditioning, cross-device correlation, and cross-user correlation. The workflows may be implemented in a clinical context, a consumer context, or a combined context, while maintaining privacy-preserving outputs.

B. Workflow 1 — Longitudinal External Monitoring With Context Conditioning

In one example workflow, a user wears an external longitudinal node for extended measurement during sleep and/or daily activity. In parallel, the user wears an upper-body context node to capture cardiac dynamics, autonomic proxies, respiratory proxies, thermal context, activity state, and wear integrity. The computing system aligns the data streams, computes signal quality indices, segments data into time windows, and generates privacy-preserving metrics for each window. The computing system conditions interpretation of intimate-region metrics using context state labels and confidence indicators, and outputs trend curves and stability measures across multiple nights or days.

C. Workflow 2 — Intravaginal Longitudinal Monitoring With Context Conditioning

In one example workflow, a user wears an intravaginal longitudinal physiology node for extended measurement. The computing system computes impedance- and dielectric-derived metrics, pH trend curves, hydration/wetness indices, pelvic activity signatures, and confidence indicators. In certain embodiments, an upper-body context node provides matched time windows, sleep proxies, thermal context, and motion gating. The system outputs longitudinal patterns and change-over-time summaries using privacy-preserving representations.

D. Workflow 3 — External Geometry Baseline Acquisition

In one example workflow, a user performs a guided scan using an external geometry measurement node. The computing system validates scan quality using IMU-derived constraints, scan speed constraints, contact confirmation, and sensor agreement. The system computes repeatable geometry metrics including length estimates, diameter profiles, shape indices, and scan confidence indicators. The geometry metrics may be stored as baseline references and used to normalize and interpret longitudinal outputs from other nodes.

E. Workflow 4 — Intraluminal Geometry and Compliance Baseline Acquisition

In one example workflow, a user performs a guided session using an intraluminal internal geometry and compliance node configured for intravaginal and/or anal use. The device applies controlled radial stimulus using an inflatable cuff, microbladder ring, constraint band, mechanical actuator, and/or pneumatic mechanism. The system records response curves using pressure transducers and contact sensing rings/bands, validates the session using motion gating and contact confirmation, and outputs privacy-preserving internal geometry and compliance metrics including compliance indices, contact stability scores, and confidence indicators. The resulting metrics may be stored as baseline references and used for normalization, longitudinal comparison, and cross-user correlation.

F. Workflow 5 — Cross-Device Correlation and Multi-Node Interpretation

In one example workflow, the computing system correlates outputs from two or more nodes, including an external longitudinal node, an intravaginal longitudinal node, an external geometry node, an intraluminal compliance node, and an upper-body context node. The system performs temporal alignment, context-tagged segmentation, and correlation scoring. The system outputs privacy-preserving correlation results including alignment scores, lag signatures, synchrony curves, stability indicators, and confidence indicators, optionally filtered to matched context windows such as low-motion segments or sleep segments.

G. Workflow 6 — Cross-User Correlation Under Consent Gating

In one example workflow, a first user and a second user authorize cross-user correlation via explicit consent and access controls. The computing system uses privacy-preserving metrics derived from one or more nodes for each user and computes cross-user correlation scores including alignment, synchrony, and stability measures. The system may compute correlation within matched context windows and may output confidence indicators and recommendations for additional data collection to improve stability. The outputs are presented without explicit anatomical imagery and without explicit anatomical labeling.

H. Variations

Workflows may be performed in different sequences and with different node subsets. Some embodiments omit certain nodes and still perform longitudinal tracking and context conditioning. Some embodiments emphasize baseline acquisition prior to longitudinal tracking. Some embodiments perform correlation locally; others perform correlation in a cloud environment with access controls. In all cases, the system may maintain privacy-preserving outputs and confidence indicators.